Reproducibility Study: Are Sparse Autoencoders Useful for Sparse Probing?

Abstract

This report audits and reproduces the paper "Are Sparse Autoencoders Useful? A Case Study in Sparse Probing" by Engels et al. (2025) which claimed that SAEs do not improve performance on downstream tasks beyond existing baseline. We applied their methods to a newer LLM - Gemma-3-4B-IT - and found results similar to that of the paper. Further experiments aimed to test the utility of SAEs on tasks they were originally designed for (Cunningham et al., 2023) also supported the authors' claims.

1 Introduction

Sparse autoencoders (SAEs) have become a popular tool for interpreting large language model activations, yet their is little evidence of them outperforming simpler methods. Engels et al. (2025) evaluate SAEs on activation probing across four regimes: standard conditions, data scarcity, label noise and class imbalance. They find that SAE probes consistently fail to beat baseline methods. We reproduce their core experiments using Gemma-3-4B-IT and extend their analysis with four additional experiments that more directly probe whether SAE latents possess the monosemanticity and concept-separability properties claimed to justify their use (Cunningham et al., 2023).

2 Methodology

We replicate Engels et al. (2025) using Gemma-3-4B-IT in place of Gemma-2-9B. We used NVIDIA L4 GPU that has 24GB VRAM to run our experiments. We use three out of four evaluation regimes (label noise is excluded due to time constraints) and a subset of the original 152 datasets (sourced from the paper's repository) due to computational constraints (full replication on the same GPU is estimated at three weeks). Under standard conditions, layers 7, 16, 24, and 32 are used and for the remaining regimes, we fix layer 24. Rather than using the full quiver-of-arrows approach, we only use logistic regression and compare that against SAE probes. This is supported by original paper's own results (confirmed via their public repository) as logistic regression is identified as the consistently dominant baseline method. We train the following kinds of SAEs for the model we use: Standard, Standard (new), JumpReLU, Gated, TopK, BatchTopK, and Matryoshka BatchTopK, across the probed layers. We also replicate the paper's three interpretability analyses: describing top single-latent probes, pruning OOD probe latents by autointerp ranking, and detecting dataset quality issues from probe firing patterns. For our extension, we add four analyses: (1) feature selectivity via Gini coefficients to test SAE monosemanticity; (2) salient neuron overlap and Jensen-Shannon concept separability by Fereidouni et al.; (3) cross-concept separability of SAE latents versus logistic regression dimensions in representation space; and (4) model stitching experiments testing transferability of each representation type to an unrelated model.

3 Results

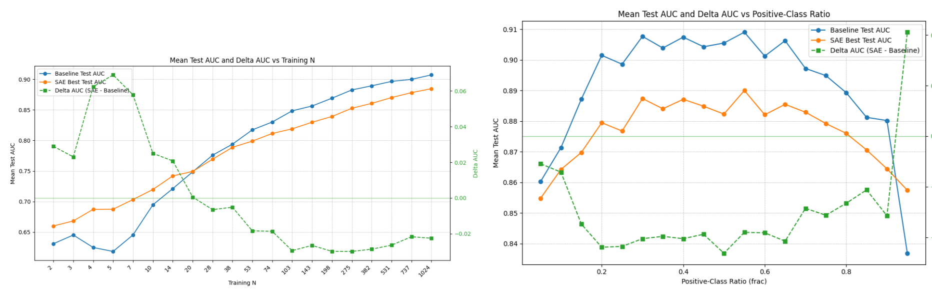
3.1 Standard Conditions

We find that the baseline method (logistic regression) is mostly marginally better than SAE probes across all the layers. From this, it can be implied that SAEs aren't bad, but they perform almost as good as logistic regression.

Subset	**Dataset Count**	**Mean Baseline AUC**	**Mean SAE AUC**	**SAE wins**	**Mean △AUC**
Overall	152	0.9253	0.9182	36 / 152	-0.0071
Layer 7	50	0.9225	0.9181	18 / 50	-0.0044
Layer 16	34	0.9368	0.9253	5 / 34	-0.0115
Layer 24	34	0.9249	0.9183	5 / 34	-0.0066
Layer 32	34	0.9183	0.9111	8 / 34	-0.0072

Table 1: Standard-condition probing results.

3.2 Regime Experiments



(a) Data Scarcity Regime Performance (b) Class imbalance Regime Performance

Figure 2: Regime Performance Results

It is observed that under data scarcity conditions, SAE probes perform well up till twenty training samples, after which Logistic Regression starts performing better. The performance gap between the two methods remains roughly the same after that point. This is a slight deviation from the original paper's results in which SAEs probes are always marginally worse than baseline methods.
For class imbalance conditions, SAE probes are worse in nearly all fractions as compared to Logistic Regression, this validates the original paper's claims where they observed the same result.

Dataset	**LogReg AUC**	**SAE AUC**	**Δ AUC**
`7_hist_fig_ispolitician`	0.597	0.679	+0.082
`67_social-security`	0.882	0.906	+0.024
`6_hist_fig_isamerican`	0.918	0.916	−0.002
`73_control-group`	0.876	0.872	−0.004
`90_glue_qnli`	0.591	0.565	−0.025
`87_glue_cola`	0.660	0.632	−0.028
`5_hist_fig_ismale`	0.966	0.901	−0.065
`66_living-room`	0.826	0.748	−0.078

Table 2: SAE vs LogReg AUC on OOD Datasets

We find that SAE probes perform better on two of out of eight OOD datasets, and even then they are not that far off from LogReg probes. This also supports the paper's claim that SAE probes do not generalize well OOD (and neither do LogReg probes).

3.3 Interpretability

Our pruning results partially replicate the paper's results: Both `66_living-room` and `7_hist_fig_ispolitician` require multiple latents rather than benefiting from pruning, while `90_glue_qnli` stays near chance, diverging from the paper's more optimistic results. This supports the paper's claims that SAEs do not generalize well on OOD data.

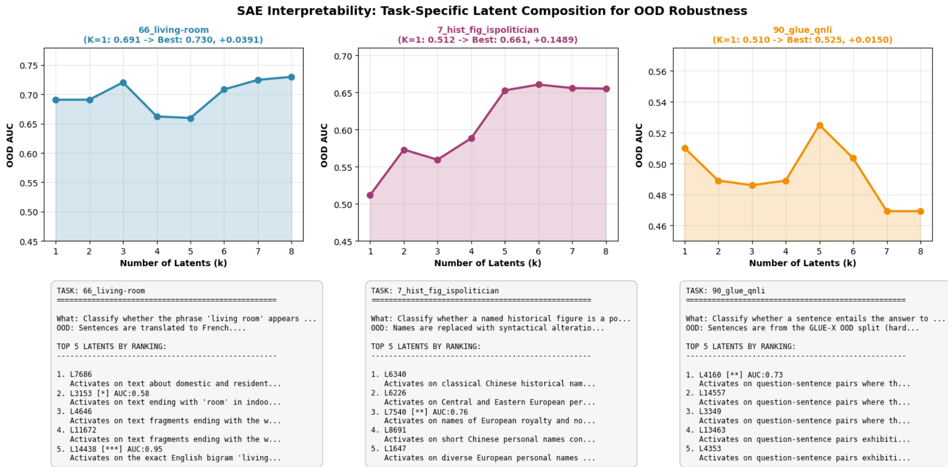


Figure 3: Interpretablity Results

3.4 Extensions

3.4.1 Feature selectivity

For each probe's top-k features, we compute the Gini coefficient of that feature's activation distribution across all training examples. A high Gini means the feature fires on a narrow subset of examples (selective). We find that SAE latents are substantially more selective than LogReg dimensions, with SAE features winning on selectivity in 82% of probes and 94% of datasets, while activation density remains similar across both methods. They also show substantially higher cross-task feature reuse than logistic regression.

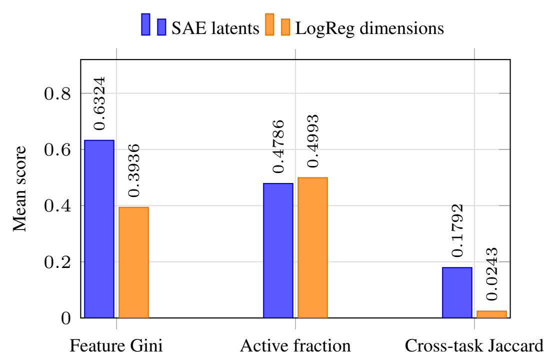


Figure 4: SAE latents vs LogReg dimensions

3.4.2 Salient neuron overlap and Jensen-Shannon concept separability (Fereidouni et al.)

We observed that SAE latents show better class-specificity than LogReg dimensions on neuron overlap, but LogReg separates classes more cleanly by JS divergence. Interestingly, even raw activations match LogReg's JS separability despite having the lowest neuron overlap, which suggests that JS divergence captures a different axis of structure than just feature selectivity.

Metric	SAE	LogReg	Raw
Salient-neuron overlap	0.6849	0.8399	0.5488
JS separability	0.528525	0.771735	0.764451
Probe AUC	0.8700	0.9109	0.9409

Table 3: Salient-neuron overlap and JS separability: SAE vs Baselines

3.4.3 Cross-concept separability in representation space

We check whether different concepts can be separated within an LLMs representation space. It is observed Cross-concept separability results are mixed with no clear winner across all metrics. Structured-sparsity variants (matryoshka, batch_topk, topk) achieve the strongest cluster separation but score worst on fragmentation, while `gated` and `p_anneal` SAEs show the opposite. No single representation dominates across all metrics.

Representation	CH	neg-DB	Silhouette	MPCD
Raw	21.17	-6.7379	-0.0660	65.326
LogReg	21.98	-9.6502	-0.2467	25.389
BatchTopK-32	40.81	-10.9717	-0.3130	13.097
Gated	27.60	-7.1076	-0.0755	90.316
JumpReLU-32	22.96	-6.8017	-0.0560	70.544
Matryoshka BatchTopK	44.14	-9.4283	-0.1861	14.841
p-anneal	22.18	-8.2729	-0.1572	109.650
Standard	22.16	-8.2523	-0.1559	109.687
Standard-new	16.57	-6.3048	-0.0639	92.200
TopK-32	32.42	-10.1828	-0.2564	12.013

Table 4: Cross-concept seperability metrics

3.4.4 Model Stitching

We find that all representations from ResNet50 trained on CIFAR-100 stitch poorly with representations from all SAEs. Among them, higher-dimensional variants (`standard`, `p_anneal`, `raw activations`) have marginally higher accuracy, while compact structured-sparsity SAEs (`batch_topk`, `topk`, `matryoshka`) and LogReg perform worst. This poor performance suggests that the LLM's internal geometry is incompatible with vision-derived concept structure.

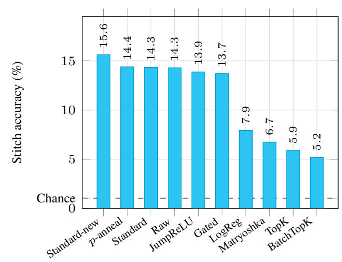


Figure 5: Model-stitching accuracy for SAEs

4 Analysis and Discussion

Our reproduction supports the original paper's central conclusion: SAE probes do not offer a reliable accuracy advantage over logistic regression across standard, class imbalance, or OOD regimes. The only exception is severe data scarcity, where SAEs' pre-structured latent space only provides a slight benefit. This benefit vanishes as sample size increases (Figure 1a). This may not generalize across models. Our interpretability analyses also indicate that SAE probes fail on OOD data (Table 2), and it is often because the latents they rely on respond to surface-level text patterns rather than the underlying concept.

Our extension results present a more nuanced picture. We find that SAE latents appear more feature-selective than logistic regression dimensions (Figure 4), which suggests they may encode more semantically coherent structure. However, SAE latents perform worse than both logistic regression and raw activations on Jensen-Shannon concept separability (Table 3), no single SAE architecture dominates on cross-concept separability (Table 4) , and compact sparse variants (TopK, BatchTopK, Matryoshka) perform near chance on model stitching, with denser variants faring slightly better likely due to higher dimensionality alone (Figure 5).

All metrics used here are proxies for interpretability rather than direct measures of it, and the original paper's conclusion stands: there is no strong evidence that SAEs provide a practical advantage over simpler methods, whether measured by classification performance or by the structural properties examined in our extension.

References

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable linguistic features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

Joshua Engels, Isaac Liao, Eric J Michaud, Wes Gurnee, and Max Tegmark. Are sparse autoencoders useful? a case study in sparse probing. *arXiv preprint arXiv:2502.16681*, 2025.

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